02_AT&T_Face_Database

November 7, 2025

Dataset Info (AT&T / ORL Face Database)

40 subjects

10 images per person

Image size: 92×112 pixels (grayscale)

Variations in facial expression, lighting, and pose (slight)

[1]:
```
from PIL import Image
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call
drive.mount("/content/drive", force_remount=True).
```

1 Using LBPH:

Adv of LBPH: Fast, easy, works well on grayscale

Built into OpenCV

No deep learning needed

Steps 1

Load all images, assign labels (1–40).

Split into training (8 images/person) and testing (2 images/person).

Use OpenCV's built-in LBPH face recognizer:

[ ]:
```
#!pip install opencv-contrib-python --quiet
#!pip install opencv-contrib-python
```

[2]:
```
import cv2
import os
import numpy as np
```

```
from sklearn.model_selection import train_test_split
```

[4]:
```
#import cv2
import os
import numpy as np
from sklearn.model_selection import train_test_split

data_path = '/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/
↪09_Face_detection/AT_T_database_of_face'
people = os.listdir(data_path)

faces, labels = [], []
for i, person in enumerate(people):
    folder = os.path.join(data_path, person)
    for img_name in os.listdir(folder):
        img = cv2.imread(os.path.join(folder, img_name), cv2.IMREAD_GRAYSCALE)
        if img is not None:
            faces.append(img)
            labels.append(i)

# Split into train and test sets
X_train, X_test, y_train, y_test = train_test_split(faces, labels, test_size=0.
↪2, random_state=42)

# Train LBPH model
model = cv2.face.LBPHFaceRecognizer_create()
model.train(list(X_train), np.array(y_train))

# Test on one sample
pred_label, confidence = model.predict(X_test[0])
print(f"Predicted: {pred_label}, Actual: {y_test[0]}, Confidence: {confidence}")
```

```
Predicted: 20, Actual: 20, Confidence: 69.09824750235762
```

[ ]:
```

```